A FIRST APPROACH TO THE AUTOMATIC DETECTION OF MARINE LITTER IN SAR IMAGES USING ARTIFICIAL INTELLIGENCE

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ABSTRACT

Remote sensing, and particularly Sinthetic Aperture Radar (SAR) is a potential tool to monitor the presence of plastic on the ocean surface, and AI algorithms can learn to find patterns in the satellite signals that lead to authomatic detection of plastic patches. The main challenge to train AI algorithms for this purpose is the lack of insitu ground truthdata that allow to reliably label the images. This paper describes a first pilot of binary pixel classification (plastic vs. non-plastic) in SAR imagery using Random Forest, GaussianNaïve Bayes and Support Vector Machines classifiers. It also describes the strategy used to generate ground truth data using Sentinel-1 and Sentinel-2 in the Balearic Islands. This ground truth consists of a total of 1794 pixels labeled as plastic and 6601 pixeles labeled as non-plastic, extracted from Sentinel-1 images.

Index Terms— SAR, AI, Marine-Litter

1. INTRODUCTION

Large amounts of litter coming from rivers, stormwater, sewage or winds ends up in the ocean, causing environmental, economic and health impacts worldwide. Moreover, most of the marine litter items degrade very slowly, increasing gradually their quantity both at sea and shores [1].

Satellite technologies have a great potential to monitor plastic waste, which would not only feed our knowledge about sources, pathways and sinks of marine litter and debris, but would also support policy making due to their evidence-based nature.

The great variety of scenarios where plastic is found in aquatic ecosystems, whether it originates from the land or at sea, of different sizes and densities, distributed in beaches, rivers and lakes, open ocean, floating on the surface, submerged along the water column and in the seabed, makes it ubiquitous and represents complex mapping requirements [2, 3].

This paper focuses on the exploitation of Synthetic Aperture Radar (SAR) images for the identification of litter signatures in marine environments. Radar technology for spaceborne remote sensing of the ocean has indisputable benefits since it provides an all-weather, all-day (independent of the solar light) wide-swath monitoring of the sea. The continuous advances in SAR coherent systems, providing high-resolution two-dimensional reflectivity images have demonstrated the suitability of this technology for marine monitoring and surveillance. The high degree of flexibility of the spaceborne SAR operation and its wide swath high-resolution imaging capabilities represents a unique and very powerful tool for ocean monitoring. Nevertheless, the particularities of marine debris, especially due to the wide variety of material types, dimensions, and interaction with the background ocean, makes its detection exploiting SAR technology very challenging. The randomness of the supporting media, where the litter is enclosed, adds complexity to the detection and identification task [3, 4].

As widely reported in the literature, the identification of a marine signature by Earth Observation technologies is an extremely relevant and challenging milestone, chased by different teams worldwide [5–10]. Recently great achievements have been reached for the detection of marine litter using optical sensor [11, 12], while a novel Floating Debris Index (FDI) [13] has been developed for the Sentinel-2 (S2) Multi-Spectral Instrument (MSI) allowing to find patches of materials on the ocean surface.

This work intends to conceptually combine different information types gathered by the radar sensor (intensity and several features from texture analysis) to enhance the detection of marine debris by a pattern recognition approach using computational intelligence techniques from SAR, with the overall objective of demonstarting the potential of this technology for this field of application. The automatic identification of marine litter in SAR images has been developed on a pixel-based approach.

To achieve this, the following points have been covered in this paper:

- Generation of ground truth from the analysis of S2 MSI images correlated with SAR acquisitions.
- SAR analysis for radar signal characterization and extraction of several features contained in the radar signal.

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- Expert pattern recognition with Artificial Intelligence (AI) to identify marine litter signatures on SAR images.

The study has been set up in the Balearic islands, an archipelago located within the Western Mediterranean Sea and currently a hotspot area for plastic marine debris accumulation situated between the two dominant currents, the Algerian and the Balearic currents [14]. Presently, it is estimated that 4.6 kg/km of plastic debris arrives at the coastline each day [15]. The accumulation of elevated concentrations of plastic debris currently poses risks to marine diversity, especially the autochthonous and endemic marine flora and fauna

2. METHODOLOGY

One of the non-negligible sources of Marine Litter is represented by meteorological extreme events. Existing literature shows the use of SAR as backscatter images of extreme events with intense plastic pollution on the ocean surface [5], [16], After big storms, tornados, or floods occurring in coastal areas, marine litter is likely introduced to the sea by either river discharging into the sea or by run-offs blown into the marine environment by winds. During the last 4 years, these events have hit more than once Mallorca where ephemeral stream systems, "torrents", become saturated and large amounts of debris that have collected in these areas during the dry season are released into the sea forming big patches of marine debris [16], and are visible with both optical and radar sensors.

2.1. Ground Truth

Due to COVID-19 restrictions, dedicated in-situ data surveying was not allowed at the time of this study. An AI model is often limited by the quality and quantity of ground truth available data, and having a large enough dataset of reliable geo-located and time-stamped marine litter observations is a common limiting requirement. To overcome the lacking of a consistent ground truth dataset of in-situ observations for this study, a dataset was obtained by applying the novel FDI [13] to the optical data from Sentinel-2 (S2) on days of reported marine litter accumulations associated to extreme events in the area of study. In short, S2 data is used as possible ground truth source to localize suspect targets in Sentinel-1 images. All dates related to extreme events where S1 and S2 acquisitions took place within a range of 4 hours are considered.
The processing flow for S2 is shown in Figure 1.

2.2. Feature extraction from SAR

The features given in input to the model, are obtained by S-1 SLC (Single Look Complex) data in IW (Interferometric Wide-swath). The SAR image processing is kept basic to maintain the maximum amount of inherent information and it is reported in the flowchart in Figure 2. It is obtained using the SNAP toolbox software provided by ESA, and it is connected

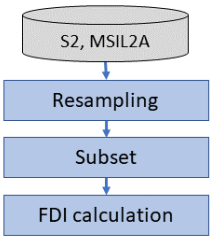


Fig. 1: S-2 processing workflow.

to the Sentinel-1 archives via the Tero-Data library by Lobelia Earth. After preprocessing steps (Applying orbit file, Removing the Thermal Noise and Calibrating), SAR data is split, debursted, and merged; then the image is geo-referenced and subsetted to reduce the data quantity to analyze. As features extracted from SAR, the intensity of the signal backscatter and the parameters derived by the Gray Level Co-occurrence Matrix (GLCM) are considered. The latter is one of the most widely used methods to compute second-order texture measures and it is useful to improve image classification accuracy. This processing is applied for both configurations (VH and VV), giving in output a total of 22 bands to feed the AI model and it does not include any type of speckle filter to avoid, as much as possible, losing any details from the backscattering signal.

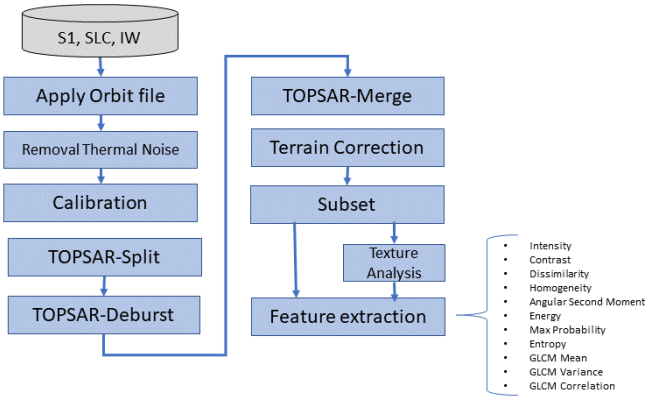


Fig. 2: SAR image processing chain.

2.3. Computational Intelligence in EO

Different AI based approaches can be applied to tackle the plastic detection on SAR images. In this paper we focused on a pixel-based binary classification:supervised binary classifiers are applied to each pixel of the image in order to decide if that pixel contains plastic or not. The three selected classification algorithms are Random Forest [17], Support Vector Machines [18] and Gaussian Naïve Bayes. The dataset obtained with the methodology explained in the previous sub-

paragraphs consists in a collection of a total of 8395 pixels from SAR images labelled as plastic (1794) and not plastic (6601). The features explained in the paragraph 2.2 are extracted for each pixel and the whole dataset is then divided in 3 data subsets:

1. The training dataset: used to train the models and adjust the fitting parameters.

2. The validation dataset: used to avoid the models overfitting to the traing dataset. The process of training the algorithms stops once the error for the validation dataset is minimized.

3. The test dataset: Once the models are trained the final performance is calculated in this independent subset so the overfitting of the model is minimal.

A PCA is applied to the 22-feature space in order to reduce dimensionality of our dataset. The seven first components of the new space were used, which explained 95% of the overall variability.

3. RESULTS

Some example images used to train and test the classifiers are shown in Figure 3. The balanced accuracy (mean of True positive and true negative rates) obtained in the binary classification is 0.68 for GNB and 0.86 for RF and SVM as shown in Table 1. This performance is calculated with the pixels within the labeled area, which does not corresponds to the whole image.
Figure 4 shows the binary detection map obtained when appliying our model to a new unlabeled image. Down-left corner presumably shows the same suspect patch detected in previous days in the neighboring areas. In this case, S-2 images of the patch where not available for the same date so we could not corroborate it with the FDI index.

Classifier	Balanced accuracy
GNB	0.68
Random Forest	0.86
Support Vector Machines	0.86

Table 1: Classification performance of the 3 models

4. CONCLUSIONS

A new methodology for the detection of marine litter from SAR technology and an AI approach has been presented, contributing to paving the way towards the use of SAR in this field.
Several studies in literature have shown the theoretical basis for the fit of radar in the detection of plastic in the ocean, but most of the achievements have been obtained with optical sensors.
In order to create a reliable ground truth for the AI model, SAR images have been labelled ("plastic" and "no-plastic") using the FDI parameter for optical imagery, considering events where both images (S1 and S2) are available. Plastic patches have been identified in SAR images. The features extracted from SAR by Intesity backscattered signal and GLCM analysis, relatively to labelled patches have been used to train the AI model.
The worst performance, in terms of balance accuracy for the labeled areas in the validation dataset, has been obtained for Gaussian Naive Bayes classifier. However, it outperforms the other methods when applied to a an unlabeled image to detect a patch, according to visual inspection of results (Figure 3). The 3 classifiers have detected the patch subject, but GNB seems to have less false positives.
There is a need for more reliable ground truth data in order to be able to improve the results, and test other AI based approaches that require larger datasets as for instance Convolutional Neural Networks. Therefore, the next steps for this study will focus on the use of in-situ validated ground truth data over a more diverse range of scenarios and locations.

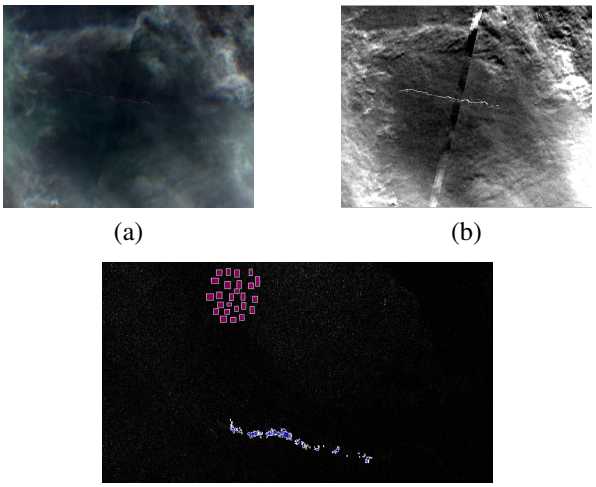


Fig. 3: Example dataset used to train and test the classifier. (a) S2 image in RGB showing plastic targets. (b) FDI applied to S2. (c) Patches in a SAR image labelled as "plastic" in blue and "no-plastic" in purple.

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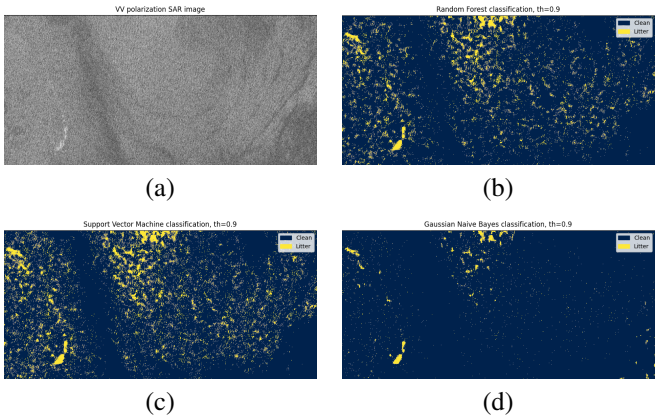


Fig. 4: Results of the pixel-based model applied to a sample image. (a) SAR image where the model was applied to. (b), (c) and (d) show in yellow the pixels with high probability of plastic with the three classification algorithm.

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